

Topic: Intersection Operation

Q. Describe the Intersection operation in Relational Algebra. What are the required conditions for applying it? Illustrate its use with a suitable example.

> **Definition to Remember**

Intersection Operation ()

- **Binary operation** operates on two tables.

2. Required Conditions (Union Compatibility)

Both relations must be **Union Compatible**:

3. Example

TOP_IN_MATH

Roll_No	Name	Grade
101	John	A
102	Emma	A
103	Michael	B

TOP_IN_SCIENCE

Roll_No	Name	Grade
101	John	B
102	Emma	A
104	Sarah	C

Query: Find students with **identical records** (same Roll_No, Name, and Grade) in both tables.

Result: Only **Emma (102, A)** her entire tuple (Roll_No=102, Name=Emma, Grade=A) is identical in both tables.

(Note: John appears in both tables but with different grades A in Math, B in Science so his full tuple is NOT identical, and he is excluded.)

> **Must-Write Points (for 10 marks)**

> 1. In

> **Quick Recall**

> ` (Int